

I will write the certificate at the level of a generic learned field h_θ , then specialize the conversion weight for score, noise, x_0 , v -prediction, and flow matching.

Thought for 24s >

Yes. The clean way is to define Q for a **generic learned object** h_θ , then specialize h .

The certificate should not be written as "score matching only." It should be written as:

$$Q_{\mathcal{P},h,s}(\rho, m, N) = Q_{0,\mathcal{P},h} + \frac{c_{\text{stat}}}{n} \int_0^B \frac{a_{\mathcal{P},h}(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{d_{\mathcal{P},h,s}(r)}{m(r)^p} dr.$$

Here:

symbol	meaning
$\mathcal{P}$	probability path, e.g. VP diffusion path, VE path, OT-CFM path
h	learned target parameterization, e.g. score, noise, velocity
s	numerical solver core
$\rho(r)$	training or schedule density
$m(r)$	sampling step density
$a_{\mathcal{P},h}(r)$	local statistical difficulty under path $\mathcal{P}$ and target h

symbol

meaning

$$d_{\mathcal{P},h,s}(r)$$

local numerical stiffness under path $\mathcal{P}$, target h , solver s

This matches the previous scheduler structure: the draft already frames Q as a certified accuracy bound decomposed into statistical and discretization terms, with the scheduler minimizing cost subject to $Q(u) \leq \varepsilon$.

Autonomous Diffusion - Autonomo...

1. Generic learned-field setup

Let $r \in [0, B]$ be intrinsic noise/path time. The path $\mathcal{P}$ defines intermediate laws

$$\mu_r^{\mathcal{P}}.$$

Let the oracle target be

$$h_{\mathcal{P}}^*(x, r).$$

A model learns

$$h_{\theta}(x, r) \approx h_{\mathcal{P}}^*(x, r).$$

Examples:

$$h = s \quad \text{score prediction,}$$

$$h = \epsilon \quad \text{noise prediction,}$$

$$h = x_0 \quad \text{data prediction,}$$

$$h = v \quad \text{velocity / flow matching.}$$

The learned object h_{θ} is converted into a physical generative field:

$$F_{\mathcal{P},h}[h_{\theta}](x, r).$$

For an SDE, F is a drift. For an ODE/flow model, F is a velocity.

Define the local physical error

$$e_F(x, r) = F_{\mathcal{P},h}[h_{\theta}](x, r) - F_{\mathcal{P},h}[h^*](x, r).$$

Assume a local linearization:

$$e_F(x, r) \approx J_{\mathcal{P},h}(r)e_h(x, r),$$

where

$$e_h(x, r) = h_\theta(x, r) - h_{\mathcal{P}}^*(x, r).$$

The matrix or scalar $J_{\mathcal{P},h}(r)$ tells how target error becomes dynamics error.

2. Generic terminal-risk stability bound

Let the terminal risk be

$$R_{\mathcal{P},h}(\theta) = \mathcal{R}(\mu_B^\theta, \mu_B^*).$$

Assume local errors propagate to terminal risk through a nonnegative weight

$$G_{\mathcal{P}}(r) \geq 0.$$

Then

$$R_{\mathcal{P},h}(\theta) - R^* \leq \int_0^B G_{\mathcal{P}}(r) \mathbb{E}_{x \sim \mu_r^{\mathcal{P}}} [\|J_{\mathcal{P},h}(r)(h_\theta - h^*)\|^2] dr + B_{\text{disc}}.$$

Define the parameterization-specific risk weight

$$w_{\mathcal{P},h}(r) = G_{\mathcal{P}}(r) \|J_{\mathcal{P},h}(r)\|_{\text{op}}^2.$$

Then

$$R_{\mathcal{P},h}(\theta) - R^* \leq \int_0^B w_{\mathcal{P},h}(r) \mathbb{E}_{\mu_r^{\mathcal{P}}} \|h_\theta - h^*\|^2 dr + B_{\text{disc}}.$$

This is the key abstraction.

3. Insert ERM

For fixed r , define local population risk

$$L_{\mathcal{P},h,r}(\theta) = \mathbb{E}_{x \sim \mu_r^{\mathcal{P}}} \|h_\theta(x, r) - h_{\mathcal{P}}^*(x, r)\|^2.$$

If training samples are allocated by density $\rho(r)$, then the effective sample size near r is $n\rho(r)$.

An ERM-style bound gives

$$L_{\mathcal{P},h,r}(\hat{\theta}_\rho) \leq b_{\mathcal{P},h}^2(r) + \frac{c_{\text{stat}} V_{\mathcal{P},h}(r)}{n\rho(r)}.$$

Here:

$$b_{\mathcal{P},h}^2(r) = \inf_{\theta \in \Theta} L_{\mathcal{P},h,r}(\theta)$$

is local approximation bias, and $V_{\mathcal{P},h}(r)$ is local statistical difficulty, such as target variance, noise level, complexity, or effective dimension.

Plug this into the terminal-risk bound:

$$R_{\mathcal{P},h}(\hat{\theta}_\rho) - R^* \leq \int_0^B w_{\mathcal{P},h}(r) b_{\mathcal{P},h}^2(r) dr + \frac{c_{\text{stat}}}{n} \int_0^B \frac{w_{\mathcal{P},h}(r) V_{\mathcal{P},h}(r)}{\rho(r)} dr + B_{\text{disc}}.$$

Define

$$Q_{0,\mathcal{P},h} = \int_0^B w_{\mathcal{P},h}(r) b_{\mathcal{P},h}^2(r) dr,$$

$$a_{\mathcal{P},h}(r) = w_{\mathcal{P},h}(r) V_{\mathcal{P},h}(r).$$

Then

$$R_{\mathcal{P},h}(\hat{\theta}_\rho) - R^* \leq Q_{0,\mathcal{P},h} + \frac{c_{\text{stat}}}{n} \int_0^B \frac{a_{\mathcal{P},h}(r)}{\rho(r)} dr + B_{\text{disc}}.$$

4. Add discretization

For solver core s of order p , write the local discretization stiffness as

$$d_{\mathcal{P},h,s}(r).$$

If $m(r)$ is the step density and

$$\int_0^B m(r) dr = N,$$

then

$$B_{\text{disc}} = c_{\text{disc}} \int_0^B \frac{d_{\mathcal{P},h,s}(r)}{m(r)^p} dr.$$

Therefore the generic certificate is:

$$R_{\mathcal{P},h,s}(\hat{\theta}_\rho, m) - R^* \leq Q_{\mathcal{P},h,s}(\rho, m, N)$$

with

$$Q_{\mathcal{P},h,s} = Q_{0,\mathcal{P},h} + \frac{c_{\text{stat}}}{n} \int_0^B \frac{a_{\mathcal{P},h}(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{d_{\mathcal{P},h,s}(r)}{m(r)^p} dr.$$

This is the general certificate.

The optimized schedule and grid are then

$$\rho_{\mathcal{P},h}^*(r) = \frac{a_{\mathcal{P},h}(r)}{\int_0^B a_{\mathcal{P},h}(q) dq},$$

$$m_{\mathcal{P},h,s}^*(r) = \frac{N d_{\mathcal{P},h,s}(r)^{1/(p+1)}}{\int_0^B d_{\mathcal{P},h,s}(q)^{1/(p+1)} dq}.$$

The optimized certificate is

$$Q_{\mathcal{P},h,s}^*(N) = Q_{0,\mathcal{P},h} + \frac{c_{\text{stat}}}{n} \left(\int_0^B a_{\mathcal{P},h}(r) dr \right)^2 + \frac{c_{\text{disc}}}{N^p} \left(\int_0^B d_{\mathcal{P},h,s}(r)^{1/(p+1)} dr \right)^{p+1}.$$

This is the frontier you invert to get minimum NFE.

Instances of h

Instance 1: score prediction

Let

$$h = s, \quad s^*(x, r) = \nabla_x \log \mu_r(x).$$

For a reverse SDE with diffusion coefficient $g(r)$, the reverse drift error is

$$e_b(x, r) = -g(r)^2 (s_\theta(x, r) - s^*(x, r)).$$

Girsanov gives path KL weight

$$\frac{1}{2} \|g(r)^{-1} e_b\|^2 = \frac{1}{2} g(r)^2 \|s_\theta - s^*\|^2.$$

So

$$w_{\mathcal{P},s}(r) = \frac{1}{2} g(r)^2.$$

Therefore

$$a_{\mathcal{P},s}(r) = \frac{1}{2} g(r)^2 V_{\mathcal{P},s}(r).$$

The certificate is

$$Q_{\mathcal{P},s} = Q_{0,\mathcal{P},s} + \frac{c_{\text{stat}}}{n} \int_0^B \frac{\frac{1}{2} g(r)^2 V_{\mathcal{P},s}(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{d_{\mathcal{P},s}(r)}{m(r)^p} dr.$$

Instance 2: noise prediction

Let

$$h = \epsilon.$$

For Gaussian corruption,

$$x_r = \alpha(r)x_0 + \sigma(r)\epsilon.$$

The conditional score satisfies

$$s^*(x_r, r) = -\frac{1}{\sigma(r)} \mathbb{E}[\epsilon \mid x_r].$$

Thus

$$s_\theta - s^* = -\frac{1}{\sigma(r)} (\epsilon_\theta - \epsilon^*).$$

So

$$\|s_\theta - s^*\|^2 = \frac{1}{\sigma(r)^2} \|\epsilon_\theta - \epsilon^*\|^2.$$

Using the same KL path weight,

$$w_{\mathcal{P},\epsilon}(r) = \frac{1}{2} \frac{g(r)^2}{\sigma(r)^2}.$$

Therefore

$$a_{\mathcal{P},\epsilon}(r) = \frac{1}{2} \frac{g(r)^2}{\sigma(r)^2} V_{\mathcal{P},\epsilon}(r).$$

So noise prediction changes the local statistical density from

$$a_{\mathcal{P},s}(r) = \frac{1}{2} g(r)^2 V_{\mathcal{P},s}(r)$$

to

$$a_{\mathcal{P},\epsilon}(r) = \frac{1}{2} \frac{g(r)^2}{\sigma(r)^2} V_{\mathcal{P},\epsilon}(r).$$

The theorem is unchanged, but the optimal schedule changes.

Instance 3: x_0 -prediction

Let

$$h = x_0.$$

From

$$x_r = \alpha(r)x_0 + \sigma(r)\epsilon,$$

we have

$$\epsilon = \frac{x_r - \alpha(r)x_0}{\sigma(r)}.$$

Thus

$$\epsilon_\theta - \epsilon^* = -\frac{\alpha(r)}{\sigma(r)}(x_{0,\theta} - x_0^*).$$

Since

$$s_\theta - s^* = -\frac{1}{\sigma(r)}(\epsilon_\theta - \epsilon^*),$$

we get

$$s_\theta - s^* = \frac{\alpha(r)}{\sigma(r)^2}(x_{0,\theta} - x_0^*).$$

Therefore

$$w_{\mathcal{P},x_0}(r) = \frac{1}{2} g(r)^2 \frac{\alpha(r)^2}{\sigma(r)^4}.$$

So

$$a_{\mathcal{P},x_0}(r) = \frac{1}{2} g(r)^2 \frac{\alpha(r)^2}{\sigma(r)^4} V_{\mathcal{P},x_0}(r).$$

This explains why x_0 -prediction has very different conditioning across noise levels.

Instance 4: v -prediction in diffusion

Diffusion v -prediction usually uses a rotation of (x_0, ϵ) , for example

$$v = \alpha(r)\epsilon - \sigma(r)x_0.$$

The conversion from v -error to score error has a path-dependent scalar factor. Write

$$s_\theta - s^* = \kappa_v(r)(v_\theta - v^*).$$

Then

$$w_{\mathcal{P},v}(r) = \frac{1}{2}g(r)^2\kappa_v(r)^2.$$

Thus

$$a_{\mathcal{P},v}(r) = \frac{1}{2}g(r)^2\kappa_v(r)^2V_{\mathcal{P},v}(r).$$

The key point is not the exact convention for v . It is that any parameterization gives a conversion factor from h -error to score/drift error:

$$h\text{-error} \rightarrow \text{score error} \rightarrow \text{terminal KL risk}.$$

That conversion factor becomes part of $a_{\mathcal{P},h}(r)$.

Instance 5: flow matching / velocity prediction

Let

$$h = v, \quad \frac{dx_r}{dr} = v^*(x_r, r).$$

The model learns

$$v_\theta(x, r) \approx v^*(x, r).$$

For deterministic flow, a natural terminal risk is Wasserstein-type:

$$R_W(\theta) = W_2^2(\mu_B^\theta, \mu_B^*).$$

Assume v^* has Lipschitz constant $L(r)$. Define terminal amplification

$$A_{\mathcal{P}}(r) = \exp \left(\int_r^B L(q) dq \right).$$

Then stability gives

$$W_2^2(\mu_B^\theta, \mu_B^*) \lesssim B \int_0^B A_{\mathcal{P}}(r)^2 \mathbb{E}_{\mu_r} \|v_\theta - v^*\|^2 dr.$$

So

$$w_{\mathcal{P},v}^{W_2}(r) = B A_{\mathcal{P}}(r)^2.$$

Therefore

$$a_{\mathcal{P},v}^{W_2}(r) = B A_{\mathcal{P}}(r)^2 V_{\mathcal{P},v}(r).$$

The certificate is

$$Q_{\mathcal{P},v}^{W_2} = Q_{0,\mathcal{P},v} + \frac{c_{\text{stat}}}{n} \int_0^B \frac{B A_{\mathcal{P}}(r)^2 V_{\mathcal{P},v}(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{d_{\mathcal{P},v,s}(r)}{m(r)^p} dr.$$

This is the flow-matching analogue of the KL score certificate. Your uploaded Wasserstein/control derivation uses the same modular form: change the terminal risk and local difficulty terms, but keep the frontier theorem unchanged.

Autonomous Diffusion - Reformul...

Summary table

h	learned target	risk weight $w_{\mathcal{P},h}(r)$	statistical density $a_{\mathcal{P},h}(r)$	
score s	$\nabla \log \mu_r$	$\frac{1}{2} g(r)^2$	$\frac{1}{2} g(r)^2 V_s(r)$	
noise ϵ	conditional noise	$\frac{1}{2} g(r)^2 / \sigma(r)^2$	$\frac{1}{2} g(r)^2 V_\epsilon(r) / \sigma(r)^2$	
data x_0	clean sample predictor	$\frac{1}{2} g(r)^2 \alpha(r)^2 / \sigma(r)^4$	$\frac{1}{2} g(r)^2 \alpha(r)^2 V_{x_0}(r) / \sigma(r)^4$	
diffusion v	rotated noise/data target	$\frac{1}{2} g(r)^2 \kappa_v(r)^2$	$\frac{1}{2} g(r)^2 \kappa_v(r)^2 V_v(r)$	
flow velocity v	transport velocity field	$BA(r)^2$	$BA(r)^2 V_v(r)$	

Final theorem form

For any path $\mathcal{P}$, learned parameterization h , and solver s , assume:

$$R_{\mathcal{P},h,s}(\hat{\theta}_\rho, m) - R^* \leq Q_{\mathcal{P},h,s}(\rho, m, N).$$

Then the accuracy-constrained scheduler is

$$(\rho^*, m^*, N^*) = \arg \min_{\rho, m, N} N \quad \text{s.t.} \quad Q_{\mathcal{P},h,s}(\rho, m, N) \leq \varepsilon.$$

Equivalently, first compute the frontier:

$$Q_{\mathcal{P},h,s}^*(N) = \min_{\rho, m} Q_{\mathcal{P},h,s}(\rho, m, N),$$

then invert:

$$N^*(\varepsilon) = \min\{N : Q_{\mathcal{P},h,s}^*(N) \leq \varepsilon\}.$$

The solution is

$$\begin{aligned} \rho^*(r) &\propto a_{\mathcal{P},h}(r), \\ m^*(r) &\propto d_{\mathcal{P},h,s}(r)^{1/(p+1)}. \end{aligned}$$

So the theorem is fully general:

Changing h changes $a_{\mathcal{P},h}(r)$.

Changing the path $\mathcal{P}$ changes $a_{\mathcal{P},h}(r)$, $d_{\mathcal{P},h,s}(r)$, and Q_0 .

Changing the solver s changes only $d_{\mathcal{P},h,s}(r)$.

That is the clean certificate hierarchy.



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Do u have close form of d(r)

There is no universal closed form for $d(r)$ independent of the sampler. But there is a closed-form template once the vector field, solver core, and terminal risk metric are fixed.

Thought for 35s >

Yes, but only after specifying:

path $\mathcal{P}$, dynamics F , solver s , risk metric R .

There is no solver-independent closed form. The general closed-form object is the **local truncation-error coefficient propagated to terminal risk**.

Let the exact continuous sampler be

$$\frac{dz_r}{dr} = F(z_r, r), \quad z_r \sim \mu_r.$$

Let $\Phi_{\Delta r}$ be the exact flow over one step, and let $\Psi_{s, \Delta r}$ be one step of solver s . If solver s has state order k , define its leading local defect by

$$\tau_s(z, r) = \lim_{\Delta r \rightarrow 0} \frac{\Psi_{s, \Delta r}(z) - \Phi_{\Delta r}(z)}{(\Delta r)^{k+1}}.$$

Then a clean closed-form template is

$$d_s(r) = \mathbb{E}_{z \sim \mu_r} [\|A_R(r, z) \tau_s(z, r)\|^q],$$

where:

$$A_R(r, z)$$

is the terminal risk amplification from local state error at time r to final risk, and q depends on the risk certificate.

For a norm-type terminal error, usually $q = 1$.

For squared Wasserstein or squared state error, usually $q = 2$.

For KL/pathwise drift-error certificates, $q = 2$ is also natural.

So the discretization term is more precisely

$$B_{\text{disc}} \lesssim c_{\text{disc}} \int_0^B \frac{d_s(r)}{m(r)^{p_{\text{risk}}}} dr,$$

where p_{risk} is the **risk decay order**, not necessarily the solver's classical state order.

If the solver state error is order k , then:

$$p_{\text{risk}} = k$$

for unsquared terminal error, but

$$p_{\text{risk}} = 2k$$

for a squared-error certificate.

This distinction matters. In your earlier notation

$$\int \frac{d_s(r)}{m(r)^p} dr,$$

the p should be interpreted as the exponent appearing in the **risk certificate**, not automatically the solver order.

Euler closed form

For Euler,

$$\Psi_{\text{Euler},h}(z, r) = z + hF(z, r).$$

The exact flow expands as

$$\Phi_h(z, r) = z + hF(z, r) + \frac{h^2}{2} (\partial_r F(z, r) + \nabla_z F(z, r) F(z, r)) + O(h^3).$$

Hence

$$\tau_{\text{Euler}}(z, r) = -\frac{1}{2} (\partial_r F(z, r) + \nabla_z F(z, r) F(z, r)),$$

up to sign, which does not matter after taking norms.

Thus

$$d_{\text{Euler}}(r) = \mathbb{E}_{z \sim \mu_r} [\|A_R(r, z) (\partial_r F(z, r) + \nabla_z F(z, r) F(z, r))\|^q].$$

Ignoring constants:

$$d_{\text{Euler}}(r) \sim \mathbb{E}_{\mu_r} [\|A_R(r, z) D_r F(z, r)\|^q]$$

where

$$D_r F = \partial_r F + \nabla F F$$

is the derivative of the vector field along the flow.

So Euler stiffness is large where the vector field changes rapidly along its own trajectory.

Heun closed form

For autonomous notation, use the extended vector field

$$F(z, r) = \begin{pmatrix} F(z, r) \\ 1 \end{pmatrix}.$$

For Heun, a second-order Runge-Kutta solver, the leading defect is third order:

$$\tau_{\text{Heun}}(z, r) = C_{\text{Heun}} \left[\nabla^2 F(F, F) - 2 \nabla F \nabla F F \right]_{z\text{-component}},$$

up to solver-dependent constants and sign.

Thus

$$d_{\text{Heun}}(r) = \mathbb{E}_{\mu_r} [\|A_R(r, z) \tau_{\text{Heun}}(z, r)\|^q].$$

Interpretation:

$$d_{\text{Heun}}(r)$$

is large where the vector field has large second directional derivatives, strong curvature, or rapidly changing Jacobian along the trajectory.

Generic Runge-Kutta / DPM / UniPC / DEIS form

For any solver s ,

$$d_s(r) = \mathbb{E}_{\mu_r} [\|A_R(r, z) \tau_s(z, r)\|^q]$$

with

$$\tau_s(z, r) = \text{leading local defect of solver } s.$$

For DPM-Solver++, UniPC, DEIS, Restart, etc., τ_s is not a simple one-line expression in F . It depends on the solver's interpolation/extrapolation formula in log-SNR or noise time. But it is still closed-form in principle:

$$\tau_s = \text{exact Taylor coefficient} - \text{solver Taylor coefficient}.$$

So for paper writing, the best definition is:

$$\tau_s(z, r) = \lim_{\Delta r \rightarrow 0} \frac{\Psi_{s, \Delta r}(z) - \Phi_{\Delta r}(z)}{(\Delta r)^{k_s+1}},$$

then define

$$d_s(r) = \mathbb{E}_{\mu_r} [\|A_R(r, z)\tau_s(z, r)\|^q].$$

That is mathematically clean and solver-general.

If the risk is KL

For reverse SDE / path-KL style certificates, local drift error is weighted by inverse diffusion covariance.

If numerical discretization induces an effective drift defect

$$\delta b_s(z, r) \approx \frac{\tau_s(z, r)}{m(r)^{k_s}},$$

then Girsanov gives

$$D_{\text{KL}}(P_{\text{exact}} \| P_{\text{num}}) \lesssim \frac{1}{2} \int_0^B \mathbb{E}_{\mu_r} \left[\Sigma(r)^{-1/2} \delta b_s(z, r)^2 \right] dr.$$

So the KL discretization density is

$$d_s^{\text{KL}}(r) = \frac{1}{2} \mathbb{E}_{\mu_r} \left[\Sigma(r)^{-1/2} \tau_s(z, r)^2 \right].$$

Then

$$B_{\text{disc}}^{\text{KL}} \lesssim \int_0^B \frac{d_s^{\text{KL}}(r)}{m(r)^{2k_s}} dr.$$

So if your notation uses

$$\int \frac{d_s(r)}{m(r)^p} dr,$$

then for this KL case,

$$p = 2k_s.$$

If the risk is terminal Wasserstein

If terminal error is propagated by flow stability,

$$A(r) = \exp \left(\int_r^B L(q) dq \right),$$

then

$$d_s^{W_2}(r) = \mathbb{E}_{\mu_r} [A(r)^2 \|\tau_s(z, r)\|^2].$$

So

$$B_{\text{disc}}^{W_2} \lesssim \int_0^B \frac{d_s^{W_2}(r)}{m(r)^{2k_s}} dr.$$

If using unsquared W_2 , then use

$$d_s^{W_2}(r) = \mathbb{E}_{\mu_r} [A(r) \|\tau_s(z, r)\|],$$

with exponent $p = k_s$.

Practical answer

Closed form exists as:

$$d_s(r) = \text{terminal amplification} \times \text{expected leading local solver defect}.$$

For Euler:

$$d_{\text{Euler}}(r) \sim \mathbb{E}_{\mu_r} [\|A_R(r, z) (\partial_r F + \nabla F F)\|^q].$$

For general solver s :

$$d_s(r) = \mathbb{E}_{\mu_r} \left[A_R(r, z) \lim_{\Delta r \rightarrow 0} \frac{\Psi_{s, \Delta r}(z) - \Phi_{\Delta r}(z)}{(\Delta r)^{k_s+1}} \right]^q.$$

Empirically, your 16-substep Heun reference estimates exactly this object: it estimates the local solver defect τ_s at each r . The empirical version is often better than writing the full symbolic derivative for modern solvers, because DPM-Solver++, UniPC, DEIS, and Restart have complicated solver-specific defect terms.





Do changing risk measure mainly change d while remaining the form of optimal m ?

Thought for 6s >

Yes, **conditional on keeping the same separable certificate form**, changing the risk measure mainly changes the **local risk-weighted stiffness** $d_R(r)$. The optimal grid form remains the same:

$$m_R^*(r) \propto d_R(r)^{1/(p+1)}.$$

But more precisely, changing the risk measure can change three objects:

$$Q_R(\rho, m, N) = Q_{0,R} + \frac{c_{\text{stat}}}{n} \int_0^B \frac{a_R(r)}{\rho(r)} dr + c_{\text{disc}} \int_0^B \frac{d_R(r)}{m(r)^p} dr.$$

changed object

why it changes

$Q_{0,R}$

different risk floors, e.g. KL floor vs Wasserstein floor

$a_R(r)$

local learning error is weighted differently by terminal risk

$d_R(r)$

local numerical defect is propagated differently into terminal risk

So the sharp answer is:

Changing risk measure changes $d_R(r)$, usually also $a_R(r)$, but not the optimizer form

The optimizer form stays because the grid subproblem remains

$$\min_m \int_0^B \frac{d_R(r)}{m(r)^p} dr \quad \text{s.t.} \quad \int_0^B m(r) dr = N.$$

The KKT condition gives

$$-\frac{p d_R(r)}{m(r)^{p+1}} + \lambda = 0,$$

so

$$m_R^*(r) = \frac{N d_R(r)^{1/(p+1)}}{\int_0^B d_R(q)^{1/(p+1)} dq}.$$

Thus the **shape law** is invariant:

$$m^*(r) \propto d(r)^{1/(p+1)}.$$

The **content** of $d(r)$ is risk-dependent.

For KL:

$$d_{\text{KL},s}(r) = \frac{1}{2} \mathbb{E}_{\mu_r} \left[\Sigma(r)^{-1/2} \tau_s(z, r)^2 \right].$$

Here the local solver defect is weighted by inverse diffusion covariance because path KL penalizes drift error through Girsanov.

For squared Wasserstein:

$$d_{W_2^2,s}(r) = \mathbb{E}_{\mu_r} \left[A(r)^2 \|\tau_s(z, r)\|^2 \right],$$

where

$$A(r) = \exp \left(\int_r^B L(q) dq \right)$$

is terminal flow amplification.

For task loss $\mathbb{E}[\ell(y, x_B)]$, a typical form is

$$d_{\ell,s}(r) = \mathbb{E}_{\mu_r} \left[\|\nabla_x \ell(y, x_B) D_z \Phi_{B \leftarrow r}(z_r) \tau_s(z_r, r)\|^2 \right].$$

So $d_R(r)$ always has the same semantic structure:

$$d_R(r) = \text{terminal risk sensitivity} \times \text{local solver defect}.$$

Changing R changes the sensitivity operator.

The same applies to $a_R(r)$:

$$a_R(r) = \text{terminal risk sensitivity} \times \text{local estimation difficulty}.$$

So KL, Wasserstein, perceptual risk, posterior risk, and constraint violation all give different a_R, d_R , but the Pareto frontier theorem remains the same as long as the certificate keeps

the decomposable form used in the uploaded scheduler notes.

Autonomous Diffusion - Autonomo...

The final theorem should say:

$$Q_R^*(N) = Q_{0,R} + \frac{c_{\text{stat}}}{n} \left(\int_0^B a_R(r) dr \right)^2 + \frac{c_{\text{disc}}}{N^p} \left(\int_0^B d_R(r)^{1/(p+1)} dr \right)^{p+1}.$$

Then

$$N_R^*(\varepsilon) = \min\{N : Q_R^*(N) \leq \varepsilon\}.$$

So: risk measure changes the profiles, not the variational optimizer form.





 Sources